

Program : Diploma in Automation and Robotics	
Course Code : 5333B	Course Title: Machine Vision
Semester: 5	Credits: 4
Course Category: Program Elective	
Periods per week: 4 (L:3 T:0 P:0)	Periods per semester: 60

Course Objectives:

- To understand the need and significance Machine Vision
- To explore the components of Machine Vision System
- To explore basics of image processing.

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Understand the vision capturing systems and its industry standards.	14	Understanding
CO2	To know about Image Representation, Matching and Inference	15	Applying
CO3	To expose the students to image feature detection and matching.	15	Applying
CO4	Apply various transform tools like frequency domain and affine transform.	14	Understanding
	Series Test	2	

CO – PO Mapping

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO1							
CO2	2	2					
CO3	2	2					
CO4		2					

3-Strongly mapped

2-Moderately mapped

1-Weakly

mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1: Understand the vision capturing systems and its industry standards.			
M1.01	What are the different types of cameras and their classification	7	Understanding
M1.02	Discuss the performance analysis of cameras.	7	Understanding
MODULE 1			
Image Acquisition, Lenses and Cameras: Pinhole camera, Gaussian Optics, Depth Field, Telemetric lenses, Lens aberrations Cameras: CCD cameras, CMOS Camera, Colour cameras: Single chip cameras, Three chip cameras, Camera performance parameters: noises, dynamic range, Camera-Computer interfaces, Digital video signals: camera link, IEEE1394, USB2.0, USB 3 vision			
CO2: To know about Image Representation, Matching and Inference			
M2.01	Explain image enhancement techniques.	8	Understanding
M2.02	Explain different image smoothing algorithms used in computer vision	7	Understanding
	Series Test – I	1	

MODULE 2			
Image Fundamentals Images, regions, piecewise contours, Image Enhancement: contrast enhancement, contrast normalization, Image Smoothing: Temporal Averaging, Mean Filter, Noise Suppression by Linear Filters, Median and Rank Filters, 2-D Fourier Transform: Continuous Fourier Transform, Discrete Fourier Transform. Geometric Transformations: Affine Transformations, Projective Transformations			
CO3: To expose the students to image feature detection and matching.			
M3.01	Explain the different image transformation techniques	8	Understanding
M3.02	Explain the different thresholding algorithms used in image processing.	7	Understanding
MODULE 3			
Image Transformations: Nearest-Neighbour Interpolation, Bilinear Interpolation, Bicubic Interpolation, Smoothing to Avoid Aliasing, Projective Image Transformations, Transformations, Image Segmentation Thresholding: Global Thresholding, Automatic Threshold Selection, Dynamic Thresholding, Variation Model, morphological operations.			
CO4: Apply various transform tools like frequency domain and affine transform.			
M4.01	Discuss edge-based approaches to segmentation	7	
M4.02	Explain the technique for texture image segmenting.	7	
	Series Test – II	1	

MODULE 4
Image Segmentation: Region Growing, Edge Based approaches to segmentation, Graph-Cut, Mean-Shift, MRFs, Texture Segmentation; Object detection.

Text / Reference

T/R	Book Title/Author
T ₁	Carsten Steger, Markus Ulrich, Christian Wiedemann, Machine Vision Algorithms and Applications, WILEY-VCH, Weinheim, 2008.
T ₂	Computer Vision: A Modern Approach, D. A. Forsyth, J. Ponce, Pearson Education, 2003.
R ₁	Rafael C. Gonzalez and Richard E. Woods, Digital Image Processing, Addition – Wesley Publishing Company, New Delhi, 2007.
R ₂	Shimon Ullman, High-Level Vision: Object recognition and Visual Cognition, A Bradford Book, USA, 2000.
R ₃	R. Patrick Goebel, ROS by Example: A Do-It-Yourself Guide to Robot Operating System -Volume I, A Pi Robot Production, 2012.